7 Hilbert Transform

Objective

After completing this chapter, the user will be able to understand

1. Continuous time Hilbert transform
2. Properties of continuous time Hilbert transform

Introduction

1. Hilbert transform is a method of separating signals based on phase selectivity, which uses phase shifts between the pertinent signals to achieve the desired separation.
2. The phase shift may be 180^0 or $\pm 90^0$.
3. For 180^0 phase shift we need an ideal transformer.

When the phase angles of a given signal are shifted by $\pm 90^0$. The resulting function of time is known as the Hilbert Transform of the signal.

The Hilbert transform has several important applications, which include the following:

1. It can be used to realise phase selectivity in the generation of a special kind of modulation known as single-sideband (SSB) modulation.
2. It provides the mathematical basis for the representation of band-pass signals.

The Hilbert transform applies to any signal that is Fourier transformable. Accordingly, it may be applied to energy signals as power signals.

Continuous Time Hilbert Transform

1. The constraint of causality of a signal implies unique relationships between the real and imaginary parts of the Fourier transform.
2. Relationships of this type between the real and imaginary parts of complex functions are known as Hilbert transform relationships.
4. We decompose x(t) into even and odd parts, i.e.,

$$x(t) = x_e(t) + x_0(t)$$

Where $x_e(t) = \frac{1}{2}\left[x(t) + x(-t)\right]$

and

$$x_0(t) = \frac{1}{2}\left[x(t) - x(-t)\right]$$

Now, if x(t) is causal, it is possible to recover x(t) from its even part $x_e(t)$ or from its odd part $x_0(t)$. Indeed, it can be easily seen that.

$$x(t) = 2x_e(t)u(t) = 2x_0(t)u(t) \quad (7.1)$$

$$x_e(t) = x_0(t)\,\text{sgn}(t) \quad (7.2)$$

and

$$x_0(t) = x_e(t)\,\text{sgn}(t) \quad (7.3)$$

Since, $x_0(t) = x_e(t)$ for $t > 0$, there is a strong relationship between $x_0(t)$ and $x_e(t)$ If $x(t)$ is absolutely integrable, the Fourier transform $X(\omega)$ exists, and

$$X(\omega) = X_R(\omega) + iX_I(\omega)$$

In addition, if x(t) is real valued and causal, the conjugate property of the Fourier transform implies that

$$x_e(t) \longleftrightarrow X_R(\omega) \text{ and } x_0(t) \longleftrightarrow jX_I(\omega)$$

The magnitude and phase responses of a causal signal are interdependent and hence cannot be specified independently.

Taking the Fourier transform of Eq. (7.2), we obtain

$$F\left[x_e(t)\right] = \frac{1}{2\pi}\left[F\left[x_0(t)\right] * F\left[\text{sgn}(t)\right]\right]$$

$$X_R(\omega) = \frac{1}{2\pi}\left[jX_I(\omega) * \frac{2}{j\omega}\right]$$

$$X_R(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} jX_I(\theta)\frac{2}{j(\omega-\theta)}d\theta$$

$$X_R(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{X_I(\theta)}{\omega-\theta}d\theta \qquad (7.4)$$

Similarly, taking the Fourier transform of Eq. (7.3), we obtain

$$F\left[x_0(t)\right] = \frac{1}{2\pi}\left[F\left[x_e(t)\right] * F\left[\text{sgn}(t)\right]\right]$$

$$jX_I(\omega) = \frac{1}{2\pi}\left[X_R(\omega) * \frac{2}{j\omega}\right]$$

$$jX_I(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X_R(\theta)\frac{2}{j(\omega-\theta)}d\theta$$

$$X_I(\omega) = -\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{X_R(\theta)}{\omega-\theta}d\theta \qquad (7.5)$$

Equations (7.4) and (7.5), which are known as the Hilbert transform pair, hold for real and imaginary parts of a causal, stable, and real signal.

Hilbert Transform Relations for Complex Signals

$x(t)$ denoting the signal and $X(\omega)$ its Fourier Transform, we require that

$$X(\omega) = 0,\ -\infty \le \omega < 0 \qquad (7.6)$$

(We could just as well that $X(\omega) = 0$ for $0 < \omega \le \infty$.) The signal $x(t)$ corresponding to $X(\omega)$ must be complex, since, if $x(t)$ were real, $X(\omega)$ would be conjugate symmetric, i.e., $X(\omega) = X^*(-\omega)$. Therefore, we express $x(t)$ as

$$x(t) = x_r(t) + jx_i(t) \qquad (7.7)$$

Where $x_r(t)$ and $x_i(t)$ are real signals. The signal $x(t)$ is called an analytical signal. If $X_r(\omega)$ and $X_i(\omega)$ denote the Fourier Transforms of the signals $x_r(t)$ and $x_i(t)$, respectively, then

$$X(\omega) = X_r(\omega) + jX_i(\omega) \qquad (7.8)$$

and it follows that

$$X_r(\omega) = \frac{1}{2}\left[X(\omega) + X^*(-\omega)\right] \qquad (7.9)$$

And

$$jX_i(\omega) = \frac{1}{2}\left[X(\omega) - X^*(-\omega)\right] \qquad (7.10)$$

Note that $X_r(\omega)$ and $X_i(\omega)$ are both complex–valued functions in general, and the complex transforms $X_r(\omega)$ and $jX_i(\omega)$ play a role similar to that played in the previous section by the even and odd parts, respectively, of a causal signal. However, $X_r(\omega)$ is conjugate symmetric and $jX_i(\omega)$ is conjugate antisymmetric, i.e.,

$$X_r(\omega) = X_r^*(-\omega) \text{ and } jX_i(\omega) = -jX_i^*(-\omega) \qquad (7.11)$$

If $X(\omega) = 0$ for $-\infty \le \omega < 0$, then there is no overlap between the portions of $X(\omega)$ and $X^*(-\omega)$. Thus, $X(\omega)$ can be recovered from either $X_r(\omega)$ or $X_i(\omega)$.

In particular,

$$X(\omega) = \begin{cases} 2X_r(\omega), & 0 \le \omega \le \infty \\ 0, & -\infty \le \omega < 0 \end{cases},$$
$$X^*(-\omega) = \begin{cases} 0, & 0 \le \omega \le \infty \\ 2X_r(\omega), & -\infty \le \omega < 0 \end{cases} \qquad (7.12)$$

And

$$X(\omega) = \begin{cases} 2jX_i(\omega), & 0 \le \omega \le \infty \\ 0, & -\infty \le \omega < 0 \end{cases},$$
$$X^*(-\omega) = \begin{cases} 0, & 0 \le \omega \le \infty \\ -2jX_i(\omega), & -\infty \le \omega < 0 \end{cases} \qquad (7.13)$$

Alternatively, from Eqs (7.12) and (7.13), we can relate $X_r(\omega)$ and $X_i(\omega)$ directly by

$$X_i(\omega) = \begin{cases} -jX_r(\omega), & 0 \le \omega \le \infty \\ jX_r(\omega) & -\infty \le \omega < 0 \end{cases} \qquad (7.14)$$

Or

$$X_i(\omega) = H(\omega)X_r(\omega) \qquad (7.15)$$

Where

$$H(\omega) = \begin{cases} -j = 1e^{-j\pi/2} & 0 < \omega \le \infty \\ j = 1e^{j\pi/2}, & -\infty \le \omega < 0 \end{cases} \qquad (7.16)$$

Or, equivalently,

$$H(\omega) = -j\,\text{sgn}(\omega) \qquad (7.17)$$

Equation (7.17) represents the frequency response of the Hilbert transformer and $x_i(t)$ is the Hilbert transform of $x_r(t)$.

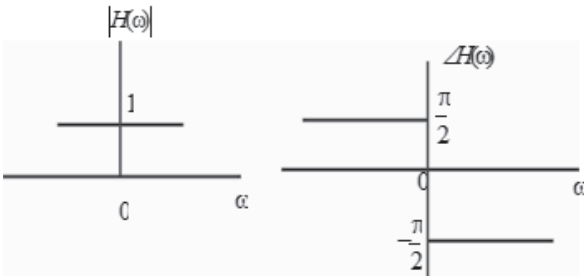


Fig. 7.1

1. Magnitude spectrum and
2. Phase response of the Hilbert transformer

Thus, $-x_r(t)$ can also be obtained from $x_i(t)$ with a 90^0 phase shifter. This is known as the inverse Hilbert transformer. The impulse response h(t) of the Hilbert transformer corresponding to the frequency response $H(\omega)$ given in Eq. (7.17) can be obtained by using the duality property of the Fourier transform. We know that

$$\text{sgn}(t) \longleftrightarrow \frac{2}{j\omega}$$

$$\frac{j}{2\pi}\text{sgn}(t) \longleftrightarrow \frac{1}{\pi\omega}$$

$$g(t) \longleftrightarrow G(\omega)$$

$$g(t) = \frac{j}{2\pi}\text{sgn}(t) \text{ and } G(\omega) = \frac{1}{\pi\omega}$$

$$g(-\omega) = \frac{j}{2\pi}\text{sgn}(-\omega)$$

$$= -\frac{j}{2\pi}\text{sgn}(\omega) \text{ and } G(t) = \frac{1}{\pi t}$$

If

$$g(t) \longleftrightarrow G(\omega)$$

Then,

$$G(t) \longleftrightarrow 2\pi g(-\omega)$$

$$\frac{1}{\pi t} \longleftrightarrow -2\pi\frac{j}{2\pi}\text{sgn}(\omega)$$

$$\frac{1}{\pi t} \longleftrightarrow -j\,\text{sgn}(\omega)$$

Therefore, the impulse response of the Hilbert transformer is given by

$$h(t) = \frac{1}{\pi t} \qquad (7.19)$$

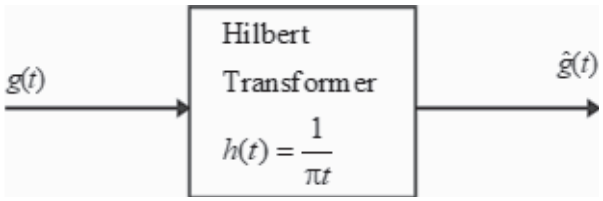


Fig. 7.2

Depicted in fig, i.e.,

$$\hat{g}(t) = g(t) * h(t)$$

$$\hat{g}(t) = g(t) * \frac{1}{\pi t}$$

$$\hat{g}(t) = \frac{1}{\pi t}\int_{-\infty}^{\infty}\frac{g(\tau)}{t-\tau}d\tau \qquad (7.20)$$

Using the convolution property of the Fourier transform, we obtain

$$F\left|\hat{g}(t)\right| = F\left|g(t)\right|F\left|h(t)\right|$$

$$\hat{G}(\omega) = G(\omega)H(\omega)$$

Using Eq. (7.17), we obtain

$$\hat{G}(\omega) = -j\,\text{sgn}(\omega)G(\omega) \qquad (7.21)$$

Equation (7.21) states that given a signal g(t), we may obtain its Hilbert transform $\hat{g}(t)$ by passing through a continuous time LTI system with frequency response $H(\omega) = -j\,\text{sgn}(\omega)$.

This system may be considered as one that produces a phase shift of -90^0 for all positive frequencies of the input signal and $+90^0$ for all negative frequencies

The inverse Hilbert transform, by means of which the original signal g(t) or $G(\omega)$ is recovered from $\hat{g}(t)$ or $\hat{G}(\omega)$, is defined by [Eq. (7.18)]

$$G(\omega) = j\,\text{sgn}(\omega)\hat{G}(\omega) \qquad (7.22)$$

Or

$$G(\omega) = -\left[-j\,\text{sgn}(\omega)\right]\hat{G}(\omega)$$

Taking the inverse Fourier transform, we obtain

$$g(t) = -\frac{1}{\pi t} * \hat{g}(t) = \hat{g}(t) * -\frac{1}{\pi t}$$

$$g(t) = -\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\hat{g}(\tau)}{t-\tau}d\tau \qquad (7.23)$$

Properties of continuous time Hilbert transform:

The signal g(t) is assumed to be real valued, which is the usual domain of application of the Hilbert transform.

1. A signal g(t) and its Hilbert transform g(t) have the same magnitude spectrum.

$$\left|\hat{G}(\omega)\right| = \left|G(\omega)\right| \tag{7.24}$$

   a) If a signal g(t) is band limited, then its Hilbert transform $\hat{g}(t)$ is also band limited.

   b) The signal g(t) and its Hilbert transform $\hat{g}(t)$ have the same energy spectral density if g(t) is an energy signal, and the same power spectral density if it is a power signal.

   c) The signal g(t) and its Hilbert transform $\hat{g}(t)$ have the same autocorrelation function.

2. If $\hat{g}(t)$ is the Hilbert transform of g(t), then the Hilbert transform of $\hat{g}(t)$ is $-g(t)$ i.e., if

$$\hat{g}(t) = HT\left[g(t)\right]$$

   Then,

$$-g(t) = HT\left[\hat{g}(t)\right] \tag{7.25}$$

3. A signal g(t) and its Hilbert transform $\hat{g}(t)$ are orthogonal, i.e., if g(t) is an energy signal, then

$$\int_{-\infty}^{\infty} g(t)\hat{g}(t)dt = 0 \tag{7.26}$$

4. An energy signal g(t) and its Hilbert transform $\hat{g}(t)$ have the same energy:

$$E_g = \int_{-\infty}^{\infty} \left|g(t)\right|^2 dt = \int_{-\infty}^{\infty} \left|\hat{g}(t)\right|^2 dt \tag{7.28}$$

   Similarly, a power signal g(t) and its Hilbert transform $\hat{g}(t)$ have the same power.

$$\lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} \left|g(t)\right|^2 dt = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} \left|\hat{g}(t)\right|^2 dt \tag{7.29}$$

7.1. Determine the Hilbert transform of $g(t) = \cos\left(\omega_c t\right)$

Solution:

Consider the given signal $g(t) = \cos\left(\omega_c t\right)$. The Fourier transform of $g(t)$ is given by

$$G(\omega) = \pi\left[\delta\left(\omega - \omega_c\right) + \delta\left(\omega + \omega_c\right)\right]$$

We know that [Eq. (7.21)],

$$\hat{G}(\omega) = -j\,\text{sgn}(\omega)G(\omega)$$

$$= -j\,\text{sgn}(\omega)\pi\left[\delta\left(\omega - \omega_c\right) + \delta\left(\omega + \omega_c\right)\right]$$

$$\hat{G}(\omega) = \frac{\pi}{j}\left[\delta\left(\omega - \omega_c\right) + \delta\left(\omega + \omega_c\right)\right]\text{sgn}(\omega)$$

It is clear that the function

$$\left[\delta\left(\omega - \omega_c\right) + \delta\left(\omega - \omega_c\right)\right]\text{sgn}(\omega)$$
$$= \left[\delta\left(\omega - \omega_c\right) - \delta\left(\omega + \omega_c\right)\right].$$

Therefore,

$$\hat{G}(\omega) = \frac{\pi}{j}\left[\delta\left(\omega - \omega_c\right) - \delta\left(\omega - \omega_c\right)\right]$$

Taking the inverse Fourier transform of the above equation, we obtain

$$\hat{g}(t) = \sin\left(\omega_c t\right)$$

Therefore

If $g(t) = \cos\left(\omega_c t\right)$, then $\hat{g}(t) = \sin\left(\omega_c t\right)$ (7.30)

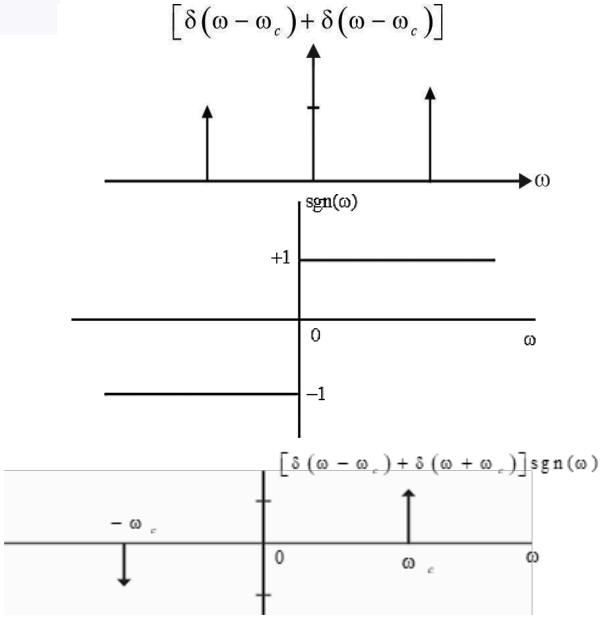


Fig. 7.3

If $g(t) = \sin\left(\omega_c t\right)$, then $g(t) = -\cos\left(\omega_c t\right)$ (7.31)

Test Your Understanding

1.

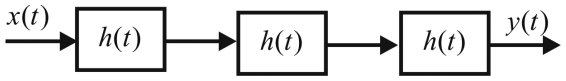


Fig. 7.4

Let $h(t) = \frac{1}{\pi t}$. If $x(t) = \cos(t)$, then $y(t)$ is given by

(A) cos (t)

(B) -cos (t)

(C) sin (t)

(D) -sin t

Solution: (D)

Pre-Envelope of Continuous Time Signals

1. Let $g(t)$ be a real valued signal. The pre-envelope of the signal g(t) is a complex-valued function defined by

$$g_+(t) = g(t) + j\hat{g}(t) \qquad (7.33)$$

Where $\hat{g}(t)$ is the Hilbert transform of the signal g(t).The pre-envelope is useful in handling band-pass signals and systems.

2. Let $G_+(\omega)$ denote the Fourier transform of $g_+(t)$.Then, by taking the Fourier transform of the Eq. (7.33), we may write

$$G_+(\omega) = G(\omega) + j\left[-j\,\mathrm{sgn}(\omega)\right]G(\omega)$$

$$= G(\omega) + \mathrm{sgn}(\omega)G(\omega)$$

$$G_+(\omega) = G(\omega)\left[1 + \mathrm{sgn}(\omega)\right]$$

$$G_+(\omega) = \begin{cases} 2G(\omega), & \omega > 0 \\ G(0), & \omega = 0 \\ 0, & \omega < 0 \end{cases} \qquad (7.34)$$

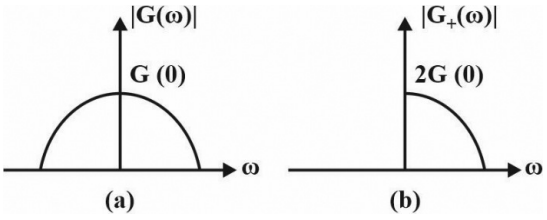


Fig. 7.5 a) Magnitude spectrum of low pass signal g(t) and b) magnitude spectrum of pre-envelope $g_+(t)$

4. Equation (7.33) defines the pre-envelope $g_+(t)$ for positive frequencies; in a similar way, we may define the pre-envelope for negative frequencies as

$$g_-(t) = g(t) - j\hat{g}(t) \qquad (7.35)$$

5. The two pre–envelopes $g_+(t)$ and $g_-(t)$ are simply the complex conjugate of each other, as shown by

$$g_-(t) = g_+^*(t) \qquad (7.36)$$

6. The spectrum of the pre-envelope $g_+(t)$ is non-zero only for positive frequencies, hence the use of a plus sign as the subscript.

7. In contrast, the spectrum of the other pre-envelope $g_-(t)$ is non-zero only for negative frequencies, as shown by the Fourier transform

$$G_-(\omega) = \begin{cases} 0, & \omega > 0 \\ G(0), & \omega = 0 \\ 2G(\omega), & \omega < 0 \end{cases}$$

7.2. Compute the Hilbert transform and pre-envelope of x(t)=sinc(2t).

Solution:

We know that,

$$\mathrm{sinc}(t) \longleftrightarrow rect\left(\frac{\omega}{2\pi}\right)$$

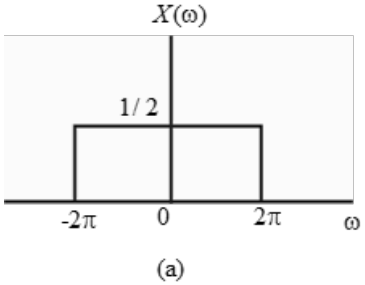


Fig. 7.6

Using the time-scaling property of the Fourier transform, we obtain

$$x(t) = \mathrm{sinc}(2t) \longleftrightarrow X(\omega) = \frac{1}{2} rect\left(\frac{\omega}{4\pi}\right)$$

The Fourier transform $X(\omega)$ is depicted in a). The Fourier transform $X(\omega)$ of the Hilbert transform $\hat{x}(t)$ is given by

b)

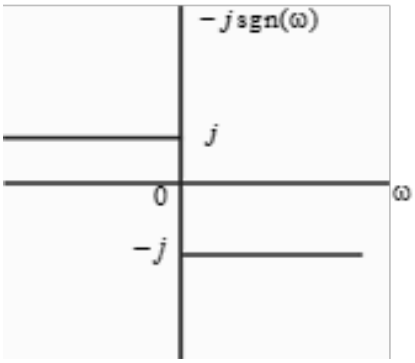


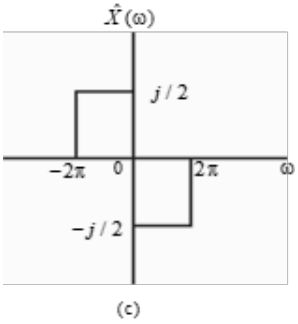
Fig. 7.7

$$\hat{X}(\omega) = -j\,\mathrm{sgn}(\omega)X(\omega)$$

$$\hat{X}(\omega) = -j\,\mathrm{sgn}(\omega)\frac{1}{2}\mathrm{rect}\left(\frac{\omega}{4\pi}\right)$$

$\hat{X}(\omega)$

shown in Fig c). Now, we shall take the inverse Fourier transform of $\hat{X}(\omega)$ to find the Hilbert transform $\hat{x}(t)$



(c)

Fig. 7.8

$$\hat{x}(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\hat{X}(\omega)e^{j\omega t}d\omega$$

$$= \frac{1}{2}\int_{-2\pi}^{0}\frac{j}{2}e^{j\omega t}d\omega + \frac{1}{2\pi}\int_{0}^{2\pi}\frac{-j}{2}e^{j\omega t}d\omega$$

The pre-envelope $x_+(t)$ of $x(t)$ is defined as

$$x_+(t) = x(t) + j\hat{x}(t)$$

$$= \sin c(2t) + \frac{1}{2\pi t}\left[1-\cos(2\pi t)\right]$$

$$= \frac{\sin(2\pi t)}{2\pi t} + \frac{1}{2\pi t}\left[1-\cos(2\pi t)\right]$$

$$x_+(t) = \frac{1}{2\pi t}\left[\sin(2\pi t) + 1 - \cos(2\pi t)\right]$$